System Design: Blockchain-Based Library Book Lending Tracker (Summative)

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Figure 1. Blockchain structure with the Merkle tree

Figure 2. Pending pool flow with prioritisation and fraud flagging

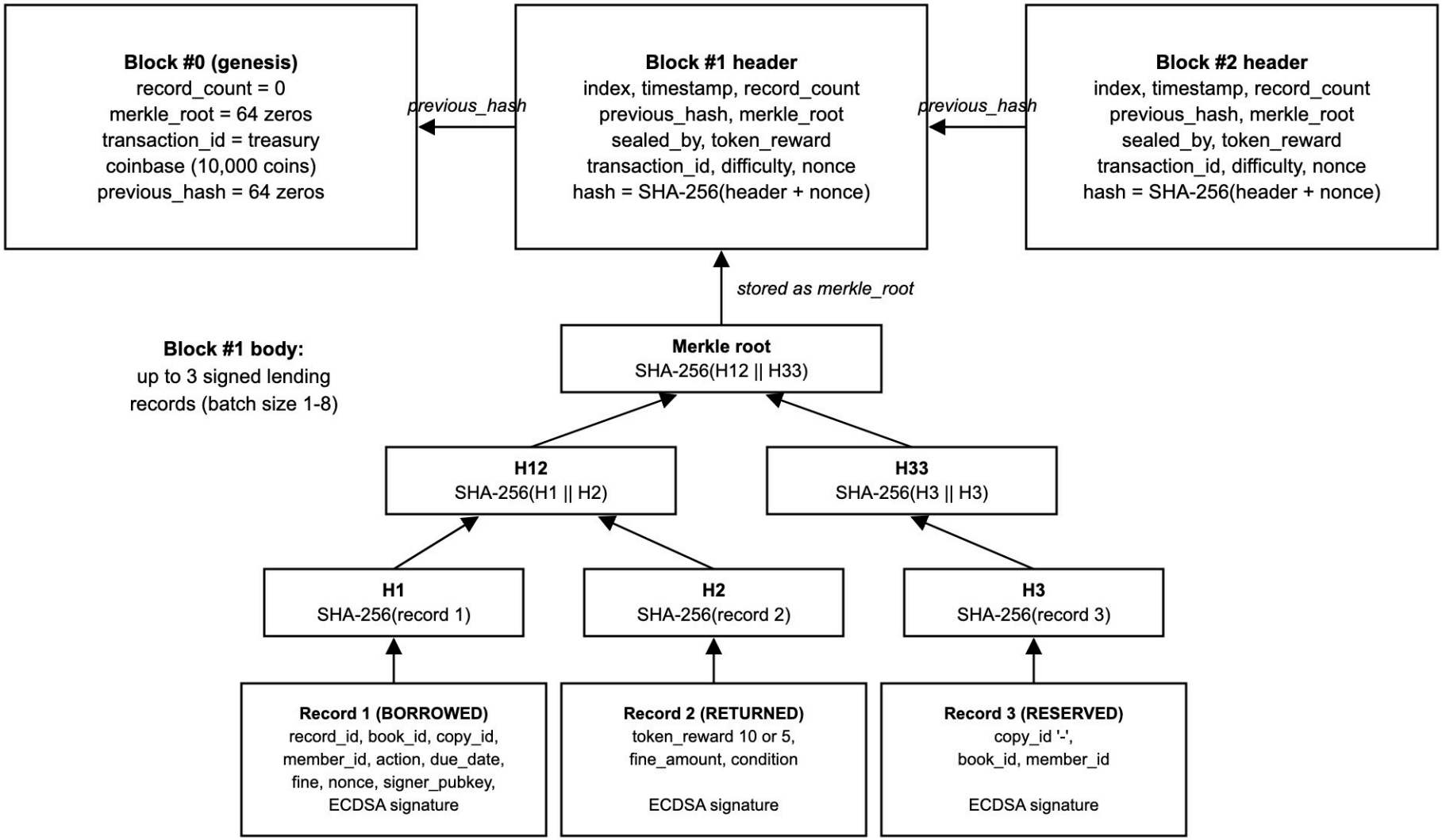
Figure 3. Copy inventory state machine and reservation queue

Figure 4. Token reward flow from return to mining confirmation to balance update

Figure 5. Mining workflow: difficulty and the solo, pool and cloud paths

Figure 6. Fraud detection pipeline from submission to review

Figure 1. Blockchain structure with the Merkle tree of each block



Odd number of nodes: the last node is paired with itself. Changing any record changes its leaf, the root, and makes validate_chain fail.

Figure 2. Pending pool flow: signing, checks, fraud flagging and prioritisation

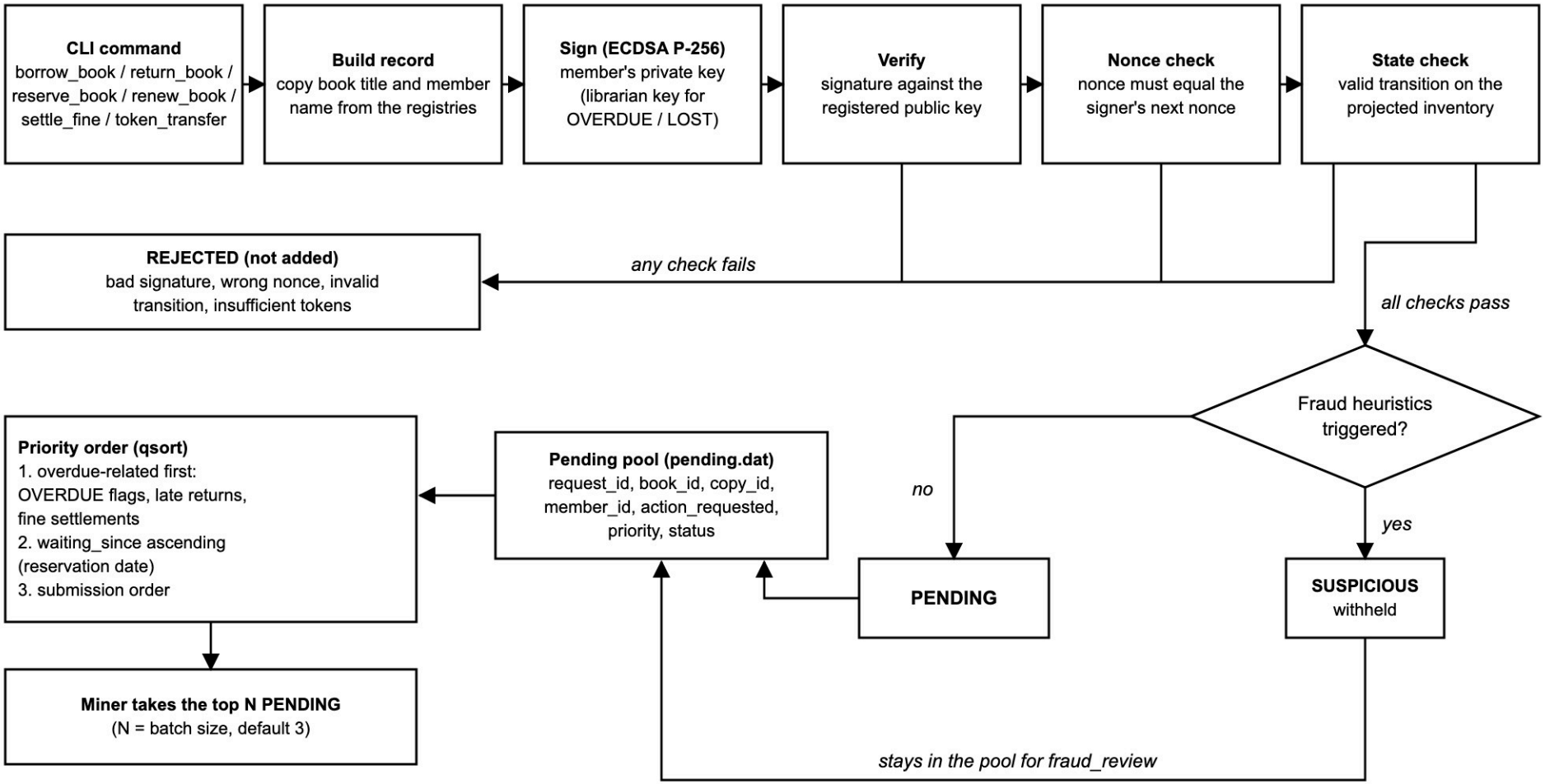
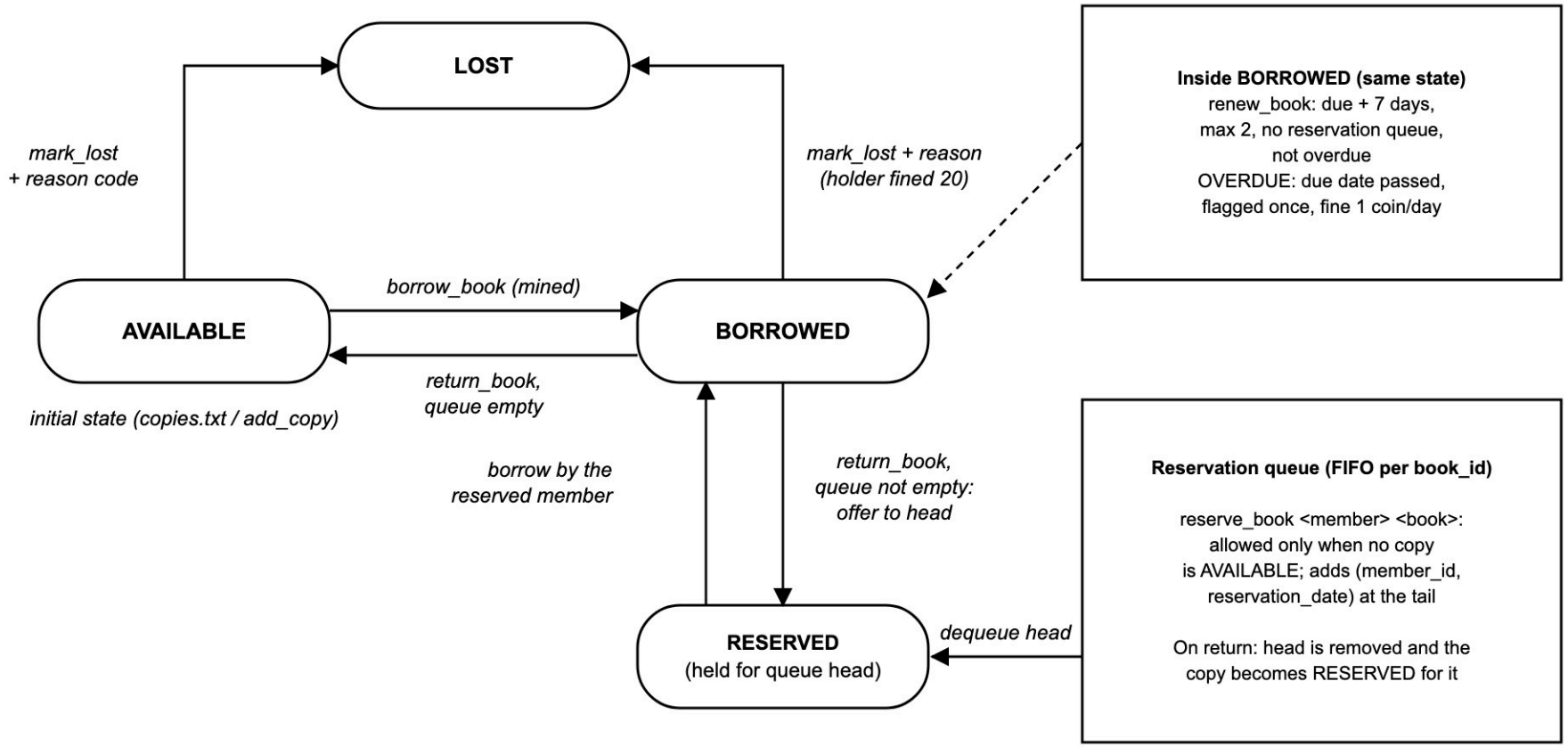
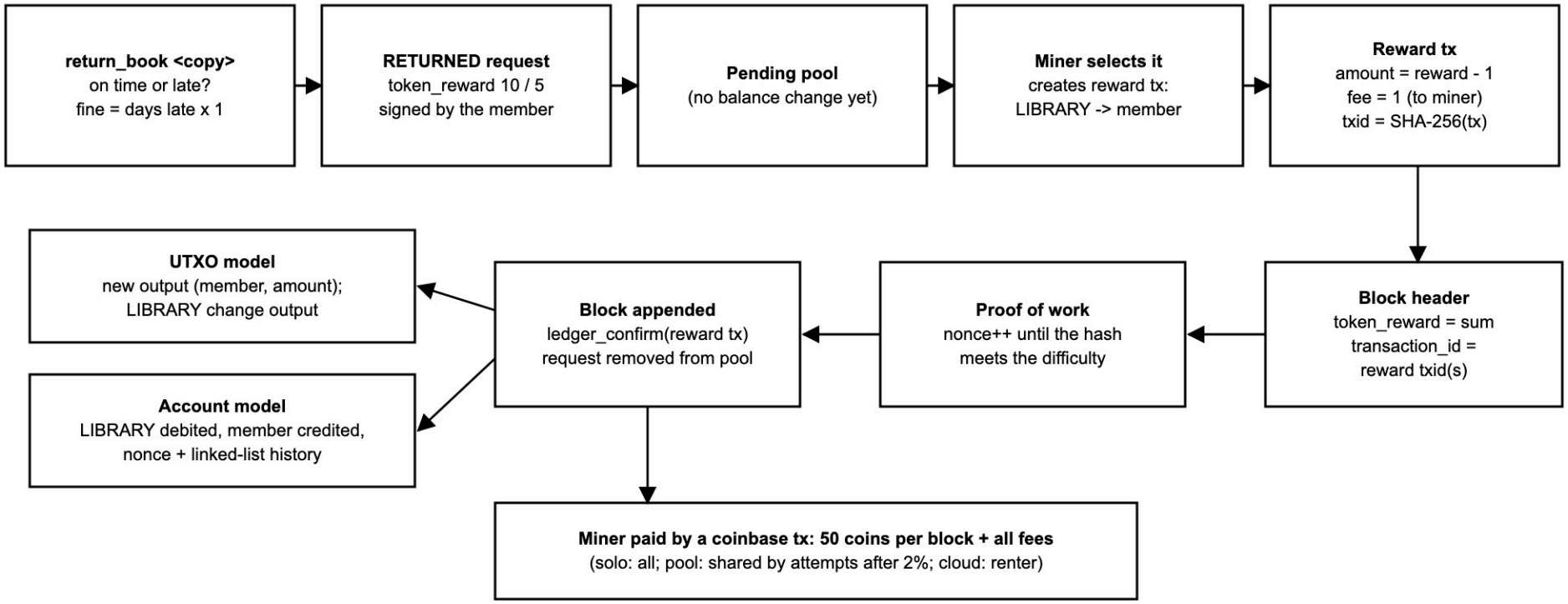


Figure 3. Copy inventory state machine and reservation queue



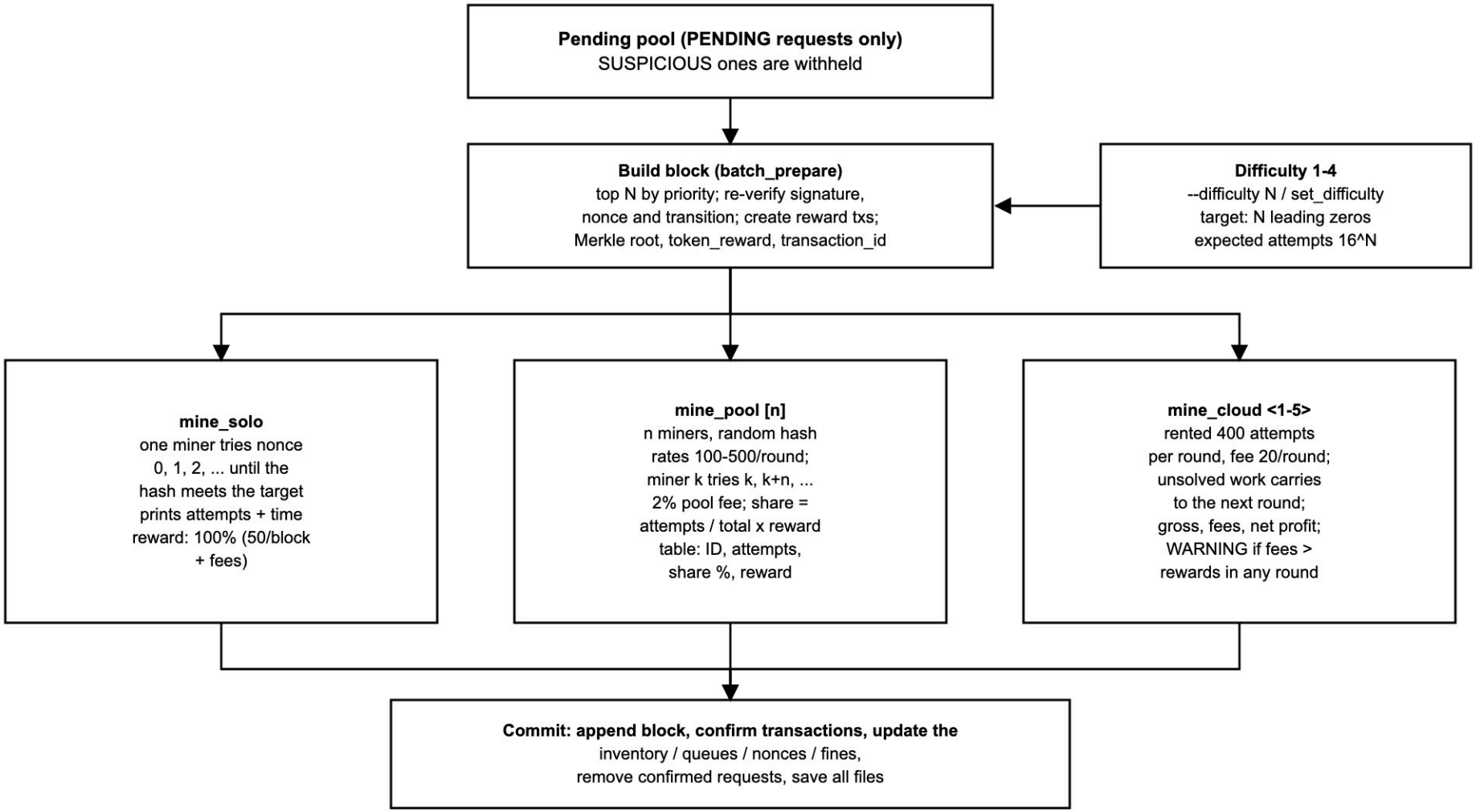
Only these transitions are accepted by state_apply(); anything else is rejected, both when a request is submitted and when the chain is replayed.

Figure 4. Token reward flow: from return to mining confirmation to balance update



A book that stays overdue and is not returned creates no reward transaction (OVERDUE record, token_reward 0).

Figure 5. Mining workflow: difficulty and the solo, pool and cloud paths



Each block stores its own difficulty, so validate_chain checks it against the difficulty it was mined at.

Figure 6. Fraud detection pipeline: from submission to review

